On my honor, I have neither given nor received any unauthorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh discipline issues.

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A-1-)

$$\begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -2\gamma & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix} \begin{matrix} -R_1+R_4=R_4 \\ -2R_1+R_3=R_3 \\ -3\alpha R_1+R_2=R_2 \end{matrix} \begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 0 & -3\alpha^2+5 & 3\gamma\alpha-2\gamma & -3\alpha^2+1 \\ 0 & -2\alpha+\beta & 2\gamma+3 & \alpha^2+2\alpha \\ 0 & 14-\alpha & \gamma-6 & 2-\alpha \end{bmatrix}$$

A-2-)

a-) $x = 5y$ is 2D plane. For volume minimum space should b 3-D.

b-) $ax + by + cz = d$

c-) $x + 4y - z = b$

$$\begin{bmatrix} 1 & 4 & -1 & b \\ 1 & -5 & 0 & 0 \end{bmatrix} \xrightarrow{-R_1 + R_2 = R_2} \begin{bmatrix} 1 & 4 & -1 & b \\ 0 & -9 & 0 & -b \end{bmatrix}$$

$$\frac{R_2}{-9} \Rightarrow R_2$$

$$\begin{bmatrix} \textcircled{1} & 4 & -1 & b \\ 0 & \textcircled{1} & 0 & 2/3 \end{bmatrix}$$

p ? f.

$$\begin{bmatrix} 1 & 4 & -1 \\ 0 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} b \\ 2/3 \end{bmatrix}$$

$c = t$

$b = 2/3$

3-) a-) If H is symmetric then $H^T = H$

If U unit vector;

$U^T . U = 1$

$H^T . H = H . H$

$H^2 = (I - 2U.U^T)(I - 2U.U^T)$

$H^2 = I^2 - 2U.U^T - 2U.U^T + 4U(U^T.U)U^T$

$H^2 = I - 4U.U^T + 4U.U^T$

$H^2 = I$

b-) H is ortagonal matrix; $H^2 = I = H . H$

c-) $H . U = (I - 2UU^T)U = U - 2UU^TU$

$= U - 2U$

$= -U$

$\lambda = -1$

A-4-) a-) $\frac{d}{dt}\begin{bmatrix} v \\ \omega \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} v \\ \omega \end{bmatrix}$

b-) $\begin{vmatrix} -1-\lambda & 1 \\ 1 & -1-\lambda \end{vmatrix} = 0$

$(-1-\lambda)(-1-\lambda) - 1 = 0$

$1+\lambda+\lambda+\lambda^2-1=0$

$\lambda^2+2\lambda=0$

$\lambda(\lambda+2)=0$

$\lambda_1 = 0$

$\lambda_2 = -2$

$y = c_1 e^{0.t} + c_2 e^{-2t}$

$y(0) = 0$